How to Approach These Questions

The assessment and interview include guesstimate and case-study prompts. Interviewers evaluate your structured thinking and stated assumptions far more than the exact final number.

General framework: clarify the question, break the problem into MECE parts (top-down or bottom-up), state assumptions explicitly, compute step by step, and sanity-check the result.

Guesstimate Questions (with structured approach)

Prompt 1. Estimate the number of smartphones sold in India in a year.

Model approach: Approach: Population ~1.4 billion; assume ~50% own a smartphone (~700M users); average replacement cycle ~3 years, so ~230M replacements/year; add first-time buyers (~30-50M/year). Estimate ~250-280 million units per year. State assumptions clearly and reason top-down.

Interviewers evaluate structure and assumptions, not the exact number.

Prompt 2. Estimate the number of cups of tea sold in a mid-sized Indian city per day.

Model approach: Approach: City population ~2 million; assume ~40% drink tea outside home (~800k people) at ~2 cups/day = 1.6M cups; add office/vendor demand. Estimate ~1.5-2 million cups/day. Segment by consumer type and justify each rate.

Break the population into segments and estimate consumption per segment.

Prompt 3. Estimate the annual revenue of a single multiplex cinema screen.

Model approach: Approach: ~5 shows/day x ~150 seats x ~40% occupancy = ~300 tickets/day; at ~Rs. 200/ticket = ~Rs. 60,000/day ticket revenue; add ~50% from food & beverages. Annual ~Rs. 60,000 x 1.5 x 350 days ~= Rs. 3.15 crore per screen.

Combine capacity, occupancy, price and ancillary revenue.

Prompt 4. Estimate how much storage (in GB) a ride-hailing app generates per day in one city.

Model approach: Approach: ~100k rides/day; each ride logs GPS pings, driver/rider IDs, payment and ratings ~ say 200 KB/ride of structured data plus location traces. 100k x 200 KB = 20 GB/day of core data; add logs/telemetry to reach a higher figure. State the data types you are counting.

Identify data types per event, then multiply by event volume.

Case Study / Analytical Prompts

Prompt 1. A retail client's online sales dropped 15% last month. How would you investigate the cause as a data analyst?

Model approach: Structured answer: (1) Confirm the data is correct (tracking, seasonality, one-off events). (2) Segment the drop by channel, region, device, product category and customer type to localise it. (3) Compare funnel metrics (traffic, conversion, average order value) to see which stage fell. (4) Check external factors (competitor promos, pricing, stockouts, site outages). (5) Form hypotheses, quantify impact of each, and recommend actions with expected uplift.

Show a MECE breakdown and tie findings to business action.

Prompt 2. A subscription business wants to reduce customer churn. What analysis and metrics would you propose?

Model approach: Structured answer: Define churn precisely and the observation window. Build a churn cohort analysis and compute churn rate, retention curves and customer lifetime value. Identify drivers using segmentation and a classification model (e.g. logistic regression / gradient boosting) with features like tenure, usage frequency, support tickets and payment issues. Rank features by importance, target high-risk segments with interventions, and measure impact via an A/B test.

Combine descriptive cohort analysis with a predictive model and an experiment.

Tredence

Guesstimates & Case Studies

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

Version 1.0 • Last Updated: July 2026

Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

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